

Nicolò Carpentieri

PhD Candidate, Systems Research Group
Technical University of Munich

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Research Interests

My interests lie in **computer architecture and systems**, particularly **memory disaggregation** at rack scale and the scale-up interconnects, like CXL and related fabrics, that make it possible. Within that space, I focus on **cache coherence protocols** for heterogeneous accelerators, including GPUs, and on automating the design and verification of the controllers that implement them.

Education

Ph.D. in Computer Science

Oct 2024 – present

Technical University of Munich

Advisor: Prof. Dr. Pramod Bhatotia

M.Sc. in Computer Engineering

2021 – 2024

Politecnico di Torino

Thesis: Designing and Evaluating Mapping of CNN Layers on an Edge-CGRA, in collaboration with the Embedded Systems Laboratory (ESL), EPFL

B.Sc. in Electronic Engineering

2018 – 2021

Politecnico di Torino

Research Experience

Visiting Student, Embedded Systems Laboratory (ESL), EPFL

2023 – 2024

Mapping of convolutional neural network algorithms onto CGRA architectures, in collaboration with DAUIN, Politecnico di Torino.

Publications

1. Decoupling Protocol, Topology, and Policy for Automated Synthesis of Coherence Controllers
Nicolò Carpentieri, Anatole Lefort, David Schall, Julian Pritzi, Pramod Bhatotia
IEEE/ACM MICRO 2026
2. vCXLGen: Automated Synthesis and Verification of CXL Bridges for Heterogeneous Architectures
Anatole Lefort, Julian Pritzi, **Nicolò Carpentieri**, David Schall, Simon Dittrich, Soham Chakraborty, Nicolai Oswald, Pramod Bhatotia
ACM ASPLOS 2026
★ **Best Paper Award**
3. C3: CXL Coherence Controllers for Heterogeneous Architectures
Anatole Lefort, David Schall, **Nicolò Carpentieri**, Julian Pritzi, Nicolai Oswald, Pramod Bhatotia
IEEE HPCA 2026

4. Performance Evaluation of Acceleration of Convolutional Layers on OpenEdgeCGRA
Nicolò Carpentieri, Juan Sapriza, Davide Schiavone, Daniele Jahier Pagliari, David Atienza,
Maurizio Martina, Alessio Burrello
ACM CF 2024

Honors and Awards

Induction in IEEE-HKN (Mu Nu Chapter), Politecnico di Torino chapter — June 2022

Teaching

Introduction to Software Engineering (EIST): 2024 – 2025, 2025 – 2026
Systems Programming: 2026 – 2027

Supervised Theses

1. Disaggregated, Vendor-Generic APUs with CXL
Victor Fritz Trost, B.Sc. Thesis, SS 2025

Technical Skills

C, C++, Python, VHDL, Verilog